

Hyper-Local Air Quality Forecasting

Problem Statement

- Current Challenges in Urban Air Quality:
 - • Limited spatial resolution of monitoring stations.
 - • Inability to provide street-level AQI forecasts.
 - • Lack of personalized health alerts for sensitive populations.

Proposed Solution: Multi-Agent System

- A decentralized architecture for intelligent forecasting:
 - • Data Agent: Real-time multi-source data ingestion.
 - • Spatial Agent: Advanced interpolation for street-level resolution.
 - • Forecast Agent: Machine Learning models (XGBoost) for predictive trends.
 - • Risk Agent: Persona-based health impact assessments.

System Architecture

- End-to-End Pipeline:
 - 1. Data Collection: Weather, Industrial, and Traffic data.
 - 2. Calibration: Correcting sensor biases for accuracy.
 - 3. Modeling: Comparing XGBoost vs. Linear Regression baselines.
 - 4. Visualization: Interactive Folium heatmaps and Plotly charts.

Key Features

- Intelligent Dashboard Functionalities:
 - • Street-Level Heatmaps: Visualizing AQI across Chennai.
 - • Personalized Alerts: Tailored advice for Elderly, Children, and Outdoor Workers.
 - • Dynamic Horizons: 6h, 12h, and 24h forecasting options.

Model Performance

- XGBoost Implementation Results:
 - • Outperformed traditional linear regression in accuracy.
 - • Successful integration of multi-source features.
 - • Real-time responsiveness for dashboard updates.

Conclusion & Future Scope

- Next Steps:
 - • Scaling to other cities across Tamil Nadu.
 - • Integrating mobile app notifications for proximity-based alerts.
 - • Incorporating additional pollutants (NO₂, SO₂, CO).